WORKED SOLUTION

STEP I 1998, Question 7

Pure • Geometric Series

STEP I 1998, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s1-q7

The key idea

Don't try to track the balance year by year. Follow each deposit separately instead: one payment of £a that sits in the bank for k years of interest grows to £ar^k, and the total is a geometric series. For the withdrawals, turn the same idea round. Each withdrawal is *paid for* by a piece of the starting balance that has been earning interest until it is taken out, so we compare values at a single moment in time.

Solution

The amount after the quiet period

Interest of $\rho\%$ multiplies the balance by $r = 1 + \frac{\rho}{100}$ at the end of each year.

Number the deposit years $1, 2, \ldots, m$. The deposit made at the beginning of year k earns interest at the end of years $k, k+1, \ldots, m$, which is $m - k + 1$ times. So at the end of year m it is worth ar^{m-k+1}. Adding over k, the balance at the end of year m is

$$\begin{aligned} a\left(r^m + r^{m-1} + \cdots + r\right) &= ar\left(1 + r + \cdots + r^{m-1}\right) \\ &= ar\,\frac{r^m - 1}{r - 1}, \end{aligned}$$

by the formula for the sum of a geometric series (we have $r \neq 1$ because $\rho > 0$).

Over the next l years nothing goes in or out, so this is multiplied by r another l times. The total at the end of the period is therefore

$$£a\,\frac{r^{l+1}(r^m - 1)}{r - 1},$$

as required. Call this amount P.

The withdrawals

The first withdrawal comes at the beginning of the year immediately after the quiet period, which is the moment the balance is P. Let B_k be the balance just after the kth withdrawal. Then

$$B_1 = P - b, \qquad B_k = rB_{k-1} - b \quad (k = 2, \ldots, n),$$

because between withdrawals the money earns one year's interest. Unwinding this recurrence,

$$\begin{aligned} B_n &= r^{n-1}P - b\left(r^{n-1} + r^{n-2} + \cdots + r + 1\right) \\ &= r^{n-1}P - b\,\frac{r^n - 1}{r - 1}. \end{aligned}$$

Ask yourself Why is there a factor r^{n-1} on P, and not r^n?

Between the first and the nth withdrawal there are only $n - 1$ year-ends, so the money earns interest $n - 1$ times. Interest paid after the last withdrawal doesn't matter, because the account is then empty. In the same way, the jth withdrawal (counting from the last, with $j = 0$ for the last one) gets multiplied by r^j before the nth withdrawal.

The account is empty after the nth withdrawal, so $B_n = 0$. This gives

$$r^{n-1}P = b\,\frac{r^n - 1}{r - 1}.$$

Substituting for P:

$$a\,\frac{r^{l+n}(r^m - 1)}{r - 1} = b\,\frac{r^n - 1}{r - 1},$$

and so

$$\boxed{\frac{a}{b} = \frac{r^n - 1}{r^{l+n}\,(r^m - 1)}}$$

An equivalent way to see this equation is to take "present values". Measured at the moment of the first withdrawal, the kth withdrawal is worth b/r^{k-1}, and these values must add up to P:

$$P = b\left(1 + r^{-1} + \cdots + r^{-(n-1)}\right).$$

Checks

- If $m = n$ the answer simplifies to $a/b = r^{-(l+n)}$, so $b = ar^{l+n}$. This makes sense. Pair the kth deposit with the kth withdrawal. They are exactly $m + l = l + n$ years apart, so each deposit pays for exactly one withdrawal.
- As $r \to 1$ (no interest), $\dfrac{r^n - 1}{r^m - 1} \to \dfrac{n}{m}$, so $a/b \to n/m$, that is, $ma = nb$. The money in equals the money out.

- For $\rho = 5$, $m = n = 10$, $l = 5$ and $a = 1000$, a year-by-year simulation gives $P \approx 16855.58$. The formula gives $a/b = 1.05^{-15} \approx 0.48102$, so $b \approx 2078.93$, and with this b the simulated account ends at exactly zero. A computer also confirms both formulas exactly, using fractions, on 500 random choices of r, l, m and n.

Common slips

- Getting r^m in place of r^{m+1} in the first sum. Because deposits are made at the *beginning* of each year, even the last deposit earns one year's interest.
- Writing $r^n P$ in place of $r^{n-1}P$. The interest at the end of the nth year comes after the account has been emptied.

Beyond STEP

This is the mathematics of annuities. Comparing payments by discounting them to a common date (present value) is the basic principle of actuarial and financial mathematics. The recurrence $B_k = rB_{k-1} - b$ is a first-order linear difference equation. Its solution, a particular constant plus a multiple of r^k, mirrors exactly the way we solve first-order linear differential equations, a parallel taken up in first-year courses on difference and differential equations. For more problems of this kind, see Siklos, *Advanced Problems in Mathematics*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.